MUHAMMAD ABBAS

BS Botany | Gold Medalist | Researcher | Aspire Institute Alumnus | Aspiring Plant Scientist
www.linkedin.com/in/mr-muhammad-abbas | orcid.org/0000-0002-9941-826
+92 3481383556 | mabbas55225@gmail.com

EDUCATION

University of Swat Swat, Pakistan
Bachelor of Science (BS) – Botany**, CGPA:** 4.00/4.00 (91.7%), **Rank:** 1st/35 September 2025

Undergraduate Thesis: *"Effect of Proline Seed Priming on Salt Stress Tolerance of Wild and Cultivated Lentils."*
Supervisor: Dr. Ahmad Ali

Relevant Coursework: Plant Physiology-I & II; Plant Biochemistry-I & II; Molecular Biology; Genetics-I & II; Mycology and Plant Pathology; Bacteriology and Virology; Research Methodology; and Biostatistics

HONORS, AWARDS, & SCHOLARSHIPS

- **University of Swat Chancellor's Gold Medal** (Expected 2028): Awarded to a student who stands 1st and has obtained a CGPA of 4.00 with A+ across the University in a Bachelor's degree program.
- **Certificate of Merit** (2026): Awarded to a student who stands 1st and has obtained a CGPA of 4.00 with A+ (90% or above marks)
- **University of Swat Merit Scholarship** (20,000 PKR per annum; 2021-2025): This scholarship is awarded to position holders, and I was able to maintain the award for all four years.
- **Laptop under Prime Minister's Youth Laptop Scheme** (2024): I was awarded this laptop from Higher Education Commission (HEC) of Pakistan based on merit and highest CGPA.
- **Islamic Studies Centre Academic Excellence Scholarship** (2012-2017): I was awarded a full school fee waiver (Grades 5-10) based on academic performance and ultimately topped my school.

RESEARCH INTERESTS

- Plant pathology and plant-microbe interactions
- Identification and characterization of disease resistance genes
- Genetic improvement of crops for biotic stress tolerance
- Plant defense responses and stress physiology
- Genome editing for disease resistance
- Sustainable strategies for crop protection

RESEARCH EXPERIENCE

Undergraduate Researcher July 2024 – August 2025
Plant Physiology Lab, Department of Botany, University of Swat
Supervisor: Dr. Ahmad Ali

- As part of my final year research project, I designed and conducted petri dish and pot experiments using a Randomized Complete Block Design (RCBD) to explore the effects of proline seed priming on germination, seedling growth, and salt stress tolerance of wild and cultivated lentils.
- Performed morpho-physiological and biochemical assays, including germination attributes, Relative Water Content (RWC), chlorophyll a, b and carotenoids quantification, proline determination, and antioxidant enzyme activities (SOD, POD) using UV-Vis Spectrophotometry.
- I learned using and trouble-shooting SmartRoot software (ImageJ) to analyze root system architecture, including root length, diameter, surface area, and volume of lentils.
- I performed statistical analysis, including Analysis of Variance (ANOVA) and Tukey's HSD all-pairwise comparisons on experimental datasets using MS Excel and analytical software Statistix 8.1.
- I presented and defended my findings at a viva-voce in front of external examiners from the Department of Biotechnology and Microbiology and was awarded a mark of 95% for this undergraduate project.

TEACHING EXPERIENCE

Tutor, Self-Employed, Swat, Pakistan Feb 2019 – July 2024

- Provided personalized one-on-one tutoring, leading to an average 30% improvement in student grades.
- Designed personalized learning plans to meet each student's needs, improving their understanding and retention of core subjects, resulting in students achieving positions in their class.
- Gained a reputation in the community for effective tutoring and consistent student success.

School Teacher, Danish Model High School, Swat, Pakistan Sept 2020 – Oct 2021

- Taught Science, Mathematics, and English to student of Grades 3 to 7 at a high school.
- Designed and delivered hands-on science lessons using practical demonstrations and everyday examples to improve student understanding and classroom participation by 40%. I was recognized as list of teachers ranked excellent by students for using effective teaching methods.

PUBLICATIONS

Manuscripts in Preparation

Malik, S., Fatima, E., **Abbas, M.**, Ayesha., Khan, D. A., Moeed, A., Shahid, A., & Ahmed, F. (2026, August 22). Oxytetracycline Trunk Injection Versus Non-Oxytetracycline Controls for the Management of Citrus Huanglongbing: A Systematic Review and Meta-Analysis. https://doi.org/10.17605/OSF.IO/UWEAF

Abbas, M., Ismail, M., Hussain, S., & Ali, A. Comparative responses of exogenously applied proline on germination and seedling physio-biochemical attributes of *Lens culinaris* under salt stress. (Target Journal: Journal of Plant Growth Regulation)

Safdar, A., Afzal, S., Saleem, A., Murtaza, M., Fatima, J., & **Abbas, M**. Comparative genomic and metabolomic analysis of aromatic medicinal plants for the discovery of novel biosynthetic gene clusters related to neurodegenerative diseases (Target Journal: Phytochemistry Reviews).

CONFERENCES/WEBINARS/WORKSHOPS ATTENDED

- International Conference on Sustainable Forest Management | University of Swat, Pakistan (2025)
- Plant Cell Focus Issue Webinar on Photosynthesis | American Society of Plant Biologists (ASPB) (2024)
- Climate Change and Phytosanitary Measures Global Webinar Series | IPPC & FAO (2024)
- 5th International Conference on Advances in Biosciences | University of Swat, Pakistan (2022)

PROFESSIONAL MEMBERSHIPS

- **General Member**, Multiomics – Global Research Community 2026-Date
- **Student Member**, American Society of Plant Biologists (ASPB) 2026 -Date
- **Student Member**, International Society for Molecular Plant-Microbe Interactions 2025 -Date

PROFESSIONAL EXPERIENCE

Meta-analysis Research Mentee | Multiomics Meta-analysis Mentorship Program Aug 2026 – Present

- Conducting a systematic review and meta-analysis on oxytetracycline trunk injection for citrus Huanglongbing management, with contributions to research-question development, PICOS eligibility criteria, and OSF registration.
- Designed database-specific search strategies using controlled vocabulary and relevant keywords, and managed reference organization and both automated and manual deduplication using EndNote.
- Conducted blinded title and abstract screening in Rayyan and full-text eligibility assessment; cross-checked peer decisions and helped resolve disagreements while maintaining records for PRISMA reporting.
- Reviewing included studies for data extraction, checking means and SD/SE/CI reporting, and identifying data that need conversion, extraction from figures, or requests to study authors before pooling.

Invigilator | Board of Intermediate & Secondary Education (BISE), Swat Mar 2026 – Apr 2026

- Officially appointed by BISE Swat as Invigilator for the Secondary School Certificate (SSC) Annual-I Examinations 2026 at Govt. Higher Secondary School Kabal, covering both Grade 9th and Grade 10th papers.
- I maintained strict examination discipline in the hall by enforcing BISE regulations, monitoring for misconduct, and ensuring fair, transparent, and orderly conduction of each paper.

Scientific Writing Intern | Scientific Writing Internship Program, Multiomics Aug 2025 – Nov 2025

- Learned how to identify research gaps, select a topic for thesis, research article, and review articles, create an outline, and select a journal based on scope, indexing, and impact.
- Gained hands-on training in research methodologies and biostatistics, with practical use of SPSS, GraphPad Prism, and R for data analysis and visualization.

Research Intern | Research Minds – NOVA Research Internship Program Feb 2025 – May 2025

- Completed a 4-month research internship, developed skills in research design and methodology; advanced literature searching using Google Scholar, PubMed, Scopus, Web of Science; literature review; academic writing; research proposal and grant writing; bibliography management using EndNote, and using AI tools for academic research.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Aspire Leaders Program — Alumnus | Aspire Institute July 2024 – Nov 2024

- Completed a cohort-based leadership and professional development program for emerging first-generation and limited-income university students and recent graduates.
- Completed training in leadership, communication, critical thinking, professional development, and global peer learning.
- Engaged with an international network of emerging leaders through leadership modules, masterclasses, cross-cultural communication, and reflective activities.

YOUNGS Training on Sustainable Development Goals — SDG-9 Ambassador Aug 2024

Ministry of Planning, Development and Special Initiatives, Islamabad, Pakistan

- Nominated by the University of Swat as SDG-9 Ambassador for "Industry, Innovation and Infrastructure".
- Completed a four-day national training focused on sustainable development goals, youth leadership, and action-oriented campus engagement.
- Collaborated with student ambassadors from universities across Pakistan to discuss SDG-focused initiatives and community-level implementation strategies.

APPLICABLE SKILLS

Plant Science Laboratory Skills

- **Physiological assays:** Relative Water Content (RWC); Chlorophyll a, b, and carotenoids
- **Biochemical assays:** Proline and antioxidant enzyme activities
- Measuring root system architecture using SmartRoot software (ImageJ)

Computational & Statistical Skills

- Microsoft Office packages (Word, Excel, PowerPoint); Google workspace (Google Docs, Sheets, Drive, Forms)
- Statistical analysis ability with MS Excel, Statistix 8.1, SPSS, GraphPad Prism, and R
- Reference Management and Bibliographic tools: EndNote and Zotero

Research & Publication

- Excellent ability with searching in advanced scientific databases: Google Scholar, PubMed, Scopus, and WoS
- Excellent research ability with knowledge of different citation and referencing styles, including APA and MPA

Language Skills

- Pashto, Urdu, and English (proficient; spoken and written)

Interpersonal Skills

- Communication skills, Intercultural competency, Critical thinking, Problem-solving, Decision making, Time management, Project management, Teamwork, Leadership

REFEREES

Dr. Ahmad Ali

Associate Professor, Department of Botany, Swat University, +92 334 5495028, draali@uswat.edu.pk

Dr. Junaid Khan

Assistant Professor, Department of Botany, Swat University, +92 314 9702373, khan.junaid@uswat.edu.pk

Dr. Muhammad Adil

Assistant Professor, Department of Botany, Swat University, +92 345 9062544, muhammad.adil@uswat.edu.pk